

Srijita Ghorai

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Profile

Machine Learning Engineer with hands-on experience in computer vision, NLP, and deep learning. Skilled in building and deploying ML models using Flask, working with transformer-based LLMs, and developing end-to-end ML pipelines. Experienced in data preprocessing, model evaluation, cloud deployment, and workflow automation. Passionate about applying AI to real-world problems through scalable and production-ready systems.

Education

2022 – Present **B. Tech in Computer Science and Engineering**
MCKV INSTITUTE OF ENGINEERING, LILUAH
Aggregate CGPA: **9.27(Till 3rd Year)**
YGPA: **9.39 (Latest)**

Professional Experience

2026/02 – Present **Machine Learning Engineer Intern**
Kyrotics Technologies Pvt. Ltd.

- Prepared multilingual training data by extracting text from documents and generating bilingual sentence pairs.
- Aligned sentence pairs using embedding similarity techniques to ensure high-quality translation datasets.
- Contributed to improving data accuracy and consistency for language translation models.

2025/06 – 2025/08 **Cloud Practitioner Trainee**
IEM LABS

- Deployed ML-enabled applications on AWS EC2 with secure IAM configurations.
- Worked with cloud storage (S3), networking (VPC), and automation using AWS CLI.
- Gained experience in containerized and cloud-ready ML deployments.

2023/06 – 2023/07 **Data Analyst Intern**
KrisKo Software Private Limited

- Analyzed real-world hospital data to extract operational and patient insights.
- Built Python-based data pipelines for cleaning, analysis, and visualization.
- Collaborated with teams to translate analytical findings into business insights.

Skills

Programming Languages

Python, Java, C

Machine Learning and Deep Learning

Scikit-learn, PyTorch, TensorFlow, Neural Networks, Model Evaluation, Data Handling, Feature Engineering

Cloud & Deployment

AWS (EC2, S3, IAM, RDS), Docker, GCP

Data Handling

Data Cleaning & Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Model Evaluation (Accuracy, Precision, Recall, F1-score, Confusion Matrix)

Automation and Workflow

ML Pipelines, API Integration, Exposure to workflow automation tools

Database Management System

SQL, MySQL, NoSQL

Version Control & Tools

Git, GitHub, Postman

Web Frameworks

Flask, Springboot, FastAPI

NLP & Generative AI:

Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Text Embeddings, Semantic Search, Hugging Face

Computer Vision

OpenCV, Image Preprocessing, Object Detection (YOLO – concepts & implementation)

Projects

SMS-Spam-Detector

- Built an SMS classification model using **TF-IDF vectorization** for text feature extraction.
- Applied **Machine Learning algorithms (Naïve Bayes / Logistic Regression / SVM)** to detect spam messages.
- Evaluated model performance using **precision, recall, F1-score, and confusion matrix**.

Password Strength Analyzer

- Developed a Python-based tool to evaluate password strength using length, character diversity, and security rules.
- Implemented logic to detect weak, medium, and strong passwords with actionable improvement suggestions.
- Applied string handling, conditional logic, and basic security concepts.
- Improved understanding of authentication security and password best practices.

Handwritten Digit Recognition using MNIST Dataset

- Implemented a handwritten digit recognition system using the MNIST dataset.
- The model is trained using a **neural network** to classify images of digits (0-9).
- The MNIST dataset contains **training set of 60,000 images** and **test set of 10,000 images**;
- Each image is 28x28 pixels, flattened into a 1D vector of 784 features.
- Performed image preprocessing and trained supervised ML models using Scikit-learn.
- Evaluated model performance using **accuracy** and **confusion matrix**

BookNest

Personalized Digital Library

- Implemented **ORM-based persistence** with **Hibernate/JPA** for managing user data in a **MySQL relational database**.
- Designed a **modular backend architecture** with **service, repository, and controller layers** for scalability and maintainability.
- Implemented **CRUD** operations for managing personalized book collections and user profiles.
- Configured **project dependencies** and build automation using Maven.

Hospital Data Analysis

- Analyzed hospital patient visit patterns and billing data using **Python, Pandas, and NumPy** to generate actionable healthcare insights.

- Identified top/bottom 20 patients based on visit frequency and billing amounts, enabling **data-driven patient segmentation** and **resource optimization**.
- Generated professional PDF reports with formatted tables and styling for healthcare administrators to support patient engagement and revenue analysis decisions.

News-Intelligence -RAG Powered News Q&A System

- Built a production-ready REST API service that answers natural language queries over a news corpus using a Retrieval-Augmented Generation (RAG) pipeline.
- Designed semantic search over news articles using vector embeddings and a vector database (Qdrant) to retrieve contextually relevant documents for each query.
- Integrated transformer-based embeddings (e.g., Universal Sentence Encoder) with an LLM (Google Gemini) to generate accurate, context-aware responses.
- Developed API endpoints with Node.js/Express and supported Docker-based deployment for scalable usage.
- Maintained conversation context with session caching using Redis and logged interaction history in PostgreSQL.